

Which Teachers Adopt AI and Which Ones Trust It With Students?

280,000 teachers · 55 education systems · OECD TALIS 2024

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Motivation

Everyone is investing in classroom AI but nobody knows who actually adopts it

- TALIS 2024 is the first cycle to ask teachers about AI, the first large-scale, cross-national snapshot (280,000 teachers, 55 country/education systems)
- Schools and ministries are investing in AI tools without knowing which teachers adopt them, or whether they use the AI at lesson planning or grading or guide student practice (Tripathi et al., 2025).
- Predominantly, the use of AI by both teachers and students shows a major shift in teaching and learning environments and the level of response among these targeted groups are of great concern



Data

OECD TALIS 2024 Teacher Survey

278,383

teacher records

630

teacher variables

55

countries / regions

3

school levels (primary • lower •
upper secondary)

Research Questions

RQ1: Which factors predict whether teachers used AI during the last 12 months?

Outcome: TT4G36 “Have you used AI in your teaching?” (yes/no) → adoption outcome

RQ2: Among teachers who used AI, who uses it directly with students?

Outcome: TT4G37 “Have you used AI in the following ways?” with 9 items → student-facing outcome

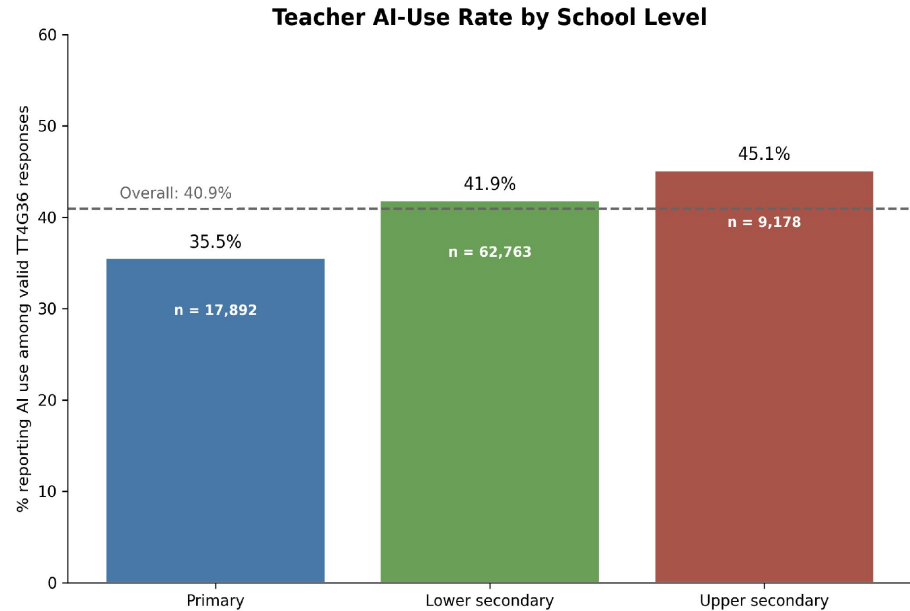
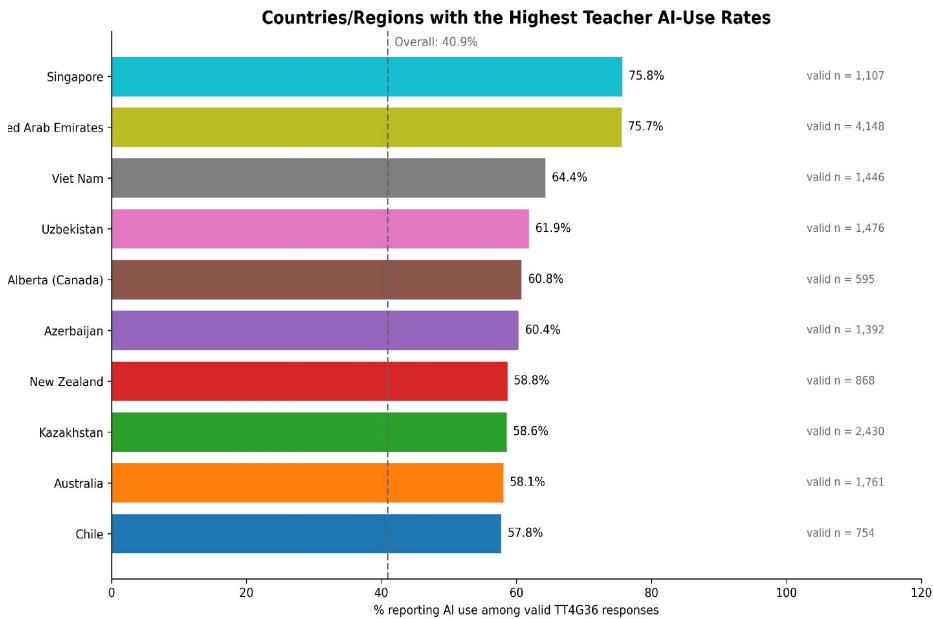
EDA RQ1 Outcome: Teachers' AI use varies by country and school level

Outcome: TT4G36 “Have you used AI in your teaching?” (yes/no)

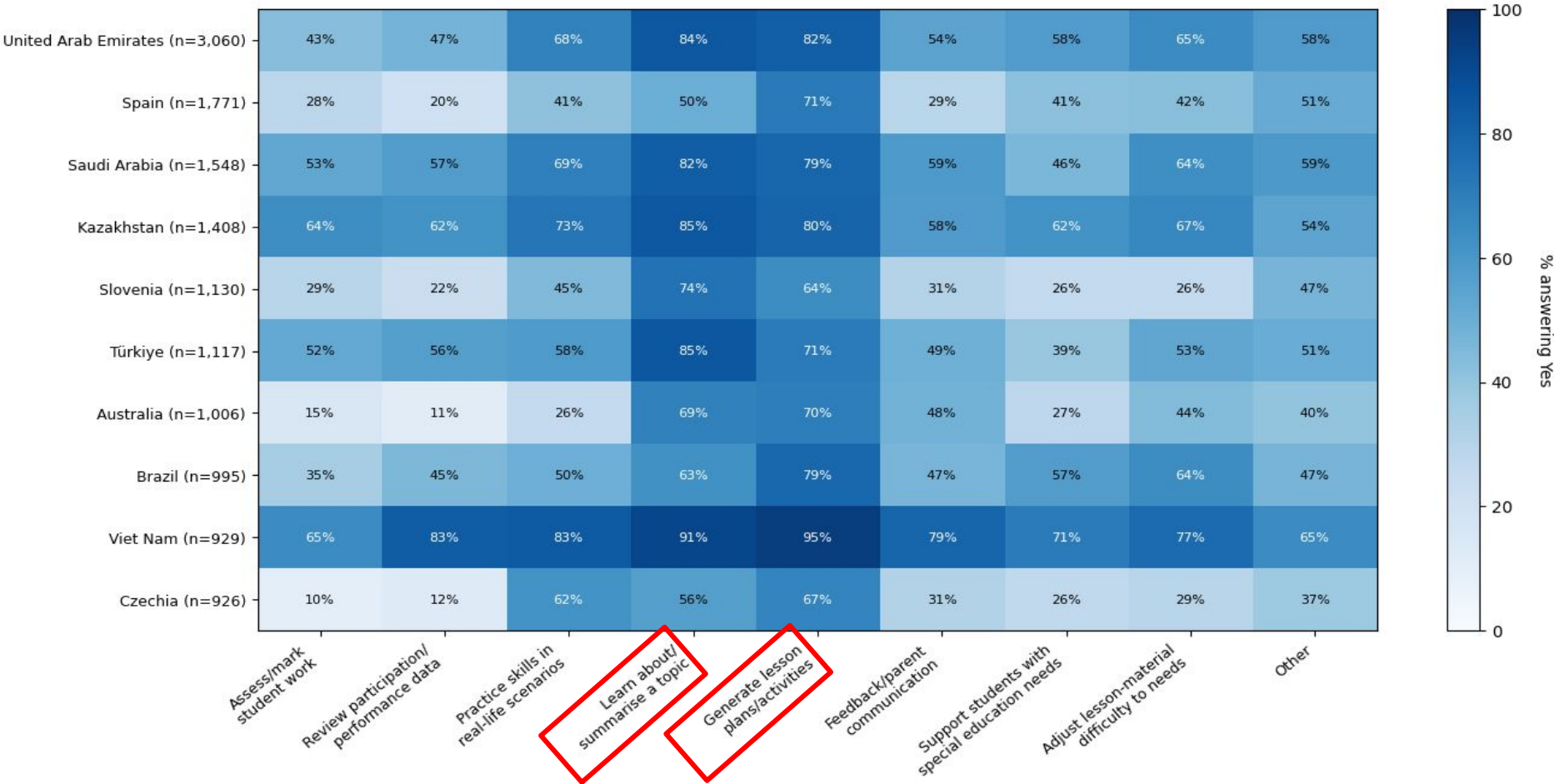
40.9%
used AI among valid TT4G36 responses

14.6-75.8%
within country / region range

35.5-45.1%
within school level range



Heatmap: AI-use purposes by top 10 country/region



RQ2 Outcome Definition

RQ2: Among teachers who used AI, which teachers used it directly with students?

Outcome: TT4G37 “Have you used AI in the following ways?” see items below → student-facing outcome

Item	Item Description	Role
TT4G37A	Assess or mark student work	student-facing (outcome)
TT4G37G	Review data on student participation or performance	student-facing (outcome)
TT4G37H	Help students practise new skills in real-life scenarios	student-facing (outcome)
TT4G37D	Support students with special education needs	ambiguous – sensitivity only
TT4G37E	Automatically adjust lesson-material difficulty	ambiguous – sensitivity only
TT4G37B	Efficiently learn about and summarise a topic	teacher-facing (descriptive)
TT4G37C	Generate lesson plans or activities	teacher-facing (descriptive)
TT4G37F	Generate text for feedback or parent communication	teacher-facing (descriptive)

References

- Tripathi, T., Sharma, S. R., Singh, V., Bhargava, P., & Raj, C. (2025, August). Teaching and learning with AI: a qualitative study on K-12 teachers' use and engagement with artificial intelligence. In *Frontiers in Education* (Vol. 10, p. 1651217). Frontiers.
- Song, Z., Qin, J., Jin, F., Cheung, W. M., & Lin, C. H. (2025). A case study of teachers' generative artificial intelligence integration processes and factors influencing them. *Teaching and Teacher Education*, 165, 105157.

EDA RQ2 Outcome: Student-facing Adoption

35,708

RQ2 valid sample

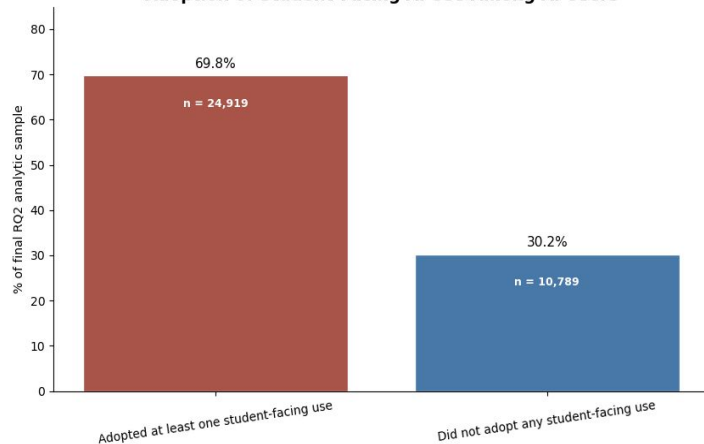
69.8%

student-facing adoption

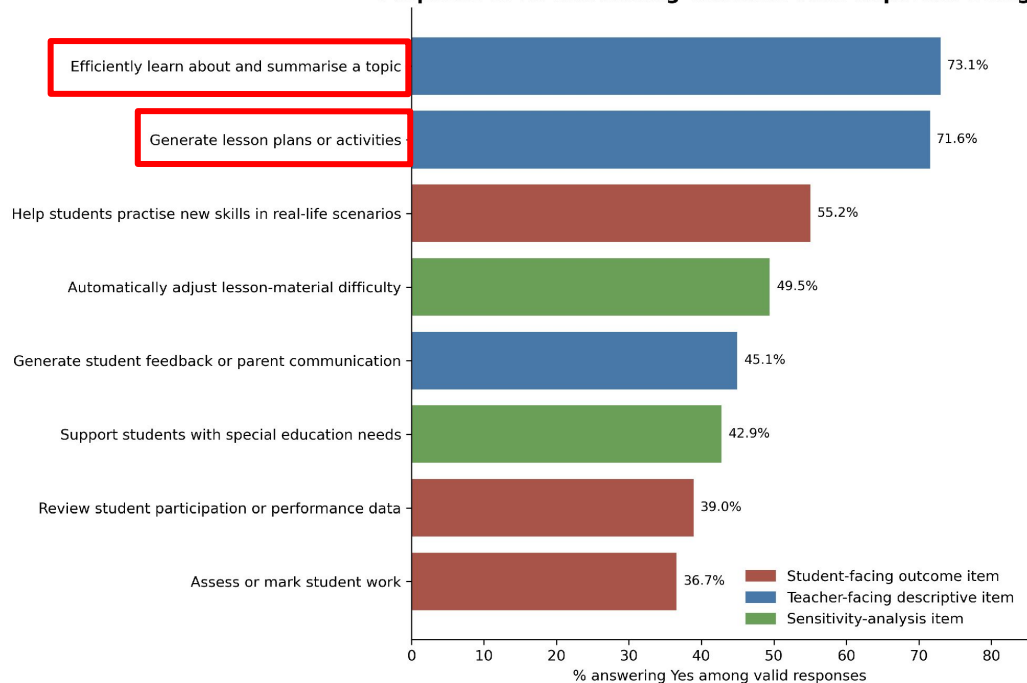
Outcome rule

Student-facing adoption = yes to at least one of TT4G37A (Assess or mark student work), TT4G37G (Review student participation/performance data), or TT4G37H (Help students practice skills in real-life scenarios).

Adoption of Student-Facing AI Use Among AI Users



Purposes of AI Use Among Teachers Who Reported Using AI



EDA: Selected Predictors and Their Relationships with AI Use

EDA retained 26 primary predictors

AI/digital preparation and professional learning: 5

- **TT4G07F** — Preparation for teaching with digital resources and tools
- **TT4G21E** — Professional learning on pedagogical use of digital tools
- **TT4G21F** — Professional learning on technical use of digital tools
- **TT4G21G** — Professional learning on AI for teaching and learning
- **TT4G27M** — Self-efficacy in supporting student learning through digital resources and tools

Collaboration and school context: 5

- **T4COLES** — Professional collaboration in lessons
- **T4EXINF** — Exchange of information and ideas among teachers
- **T40PPART** — Opportunities to participate in school decisions
- **T4STUD** — Teacher–student relations
- **T4TLEAD** — Teacher leadership

Job stress and satisfaction: 4

- **T4CHFAT** — Change fatigue
- **T4J0BSAT** — Overall job satisfaction
- **T4J0YTCH** — Joy of teaching
- **T4WLOADT** — Workload stress

Teacher background: 7

- **T4TAGEGR** — Teacher age group
- **T4TEMPWH** — Full-time versus part-time employment
- **T4THEDAT** — Highest formal education level
- **T4TNSCH** — Number of schools at which the teacher works
- **T4TYEXPTT** — Total teaching experience
- **TT4G08** — Teaching was the first career choice
- **TT4G09** — Permanent versus fixed-term employment

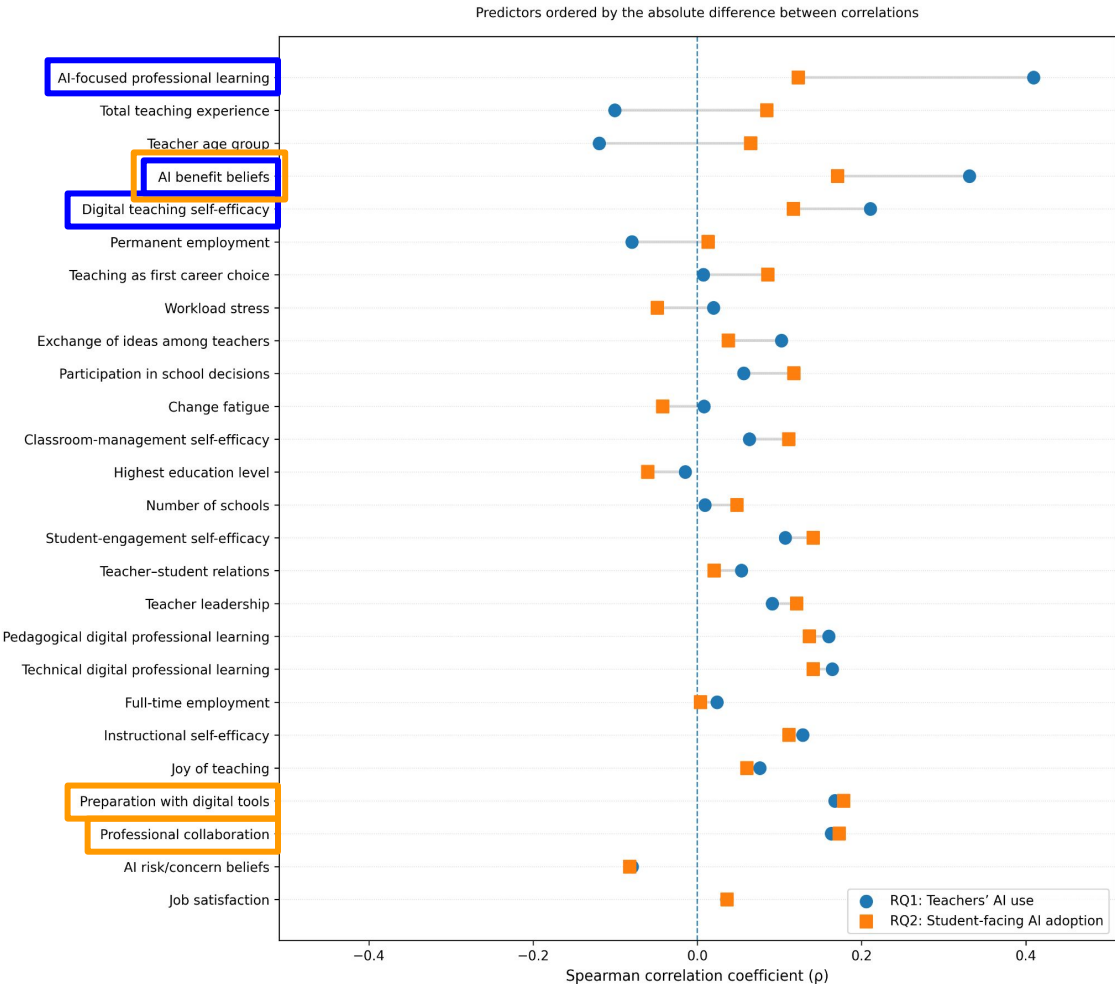
Teacher self-efficacy: 3

- **T4SECLS** — Classroom-management self-efficacy
- **T4SEENG** — Student-engagement self-efficacy
- **T4SEINS** — Instructional self-efficacy

AI-belief composites: 2

- **ai_benefit_beliefs_mean** — Higher values indicate stronger agreement with AI benefit statements
- **ai_risk_concern_beliefs_mean** — Higher values indicate stronger agreement with AI risk or concern statements

RQ1 and RQ2 Predictor Correlations



Model Setup for Two Questions

$y \sim \text{AI beliefs} + \text{AI training} + \text{teacher traits} + \text{country}$

Question 1: Adoption

n = 75,817

41.4% use AI (split 41.4 / 41.2 - balanced)

Question 2: Student Facing Users

n = 30,689 AI users after complete case

70% base rate (report ROC-AUC + PR-AUC)

FEATURES

- 18 teacher features
- 2 belief composites (benefit / risk)
- "don't know" → 2.5 midpoint

SPLIT & LEAKAGE

- School-grouped 70/30 — schools never straddle train/test, balanced (all |SMD| < 0.03)
- Pipeline fit on train only
- Null test, 20 shuffles: P1 0.504 ± 0.025 · P2 0.497 ± 0.023 ($\approx$ chance)

MODEL SELECTION

best of 7 models (grouped 5-fold CV)

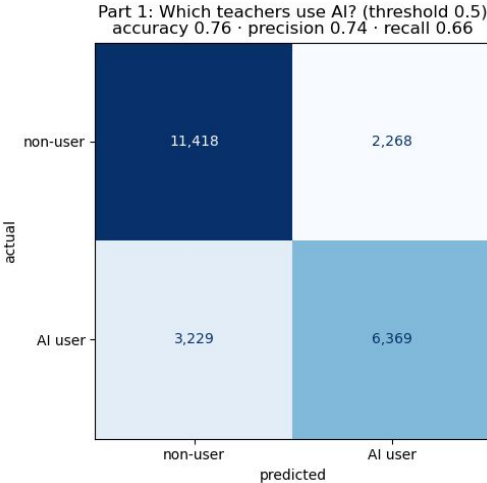
ROC-AUC + DeLong test: are AUC gaps real or noise?

Model Selection for RQ1 and RQ2

RQ1

Model Comparison — grouped 5-fold CV

Model	CV AUC	SD
GradientBoosting	0.831	±0.003
LightGBM	0.830	±0.003
RandomForest	0.825	±0.003
ElasticNet Logistic	0.815	±0.003
Lasso (L1) Logistic	0.815	±0.003
LogisticRegression	0.815	±0.003
XGBoost	0.815	±0.004

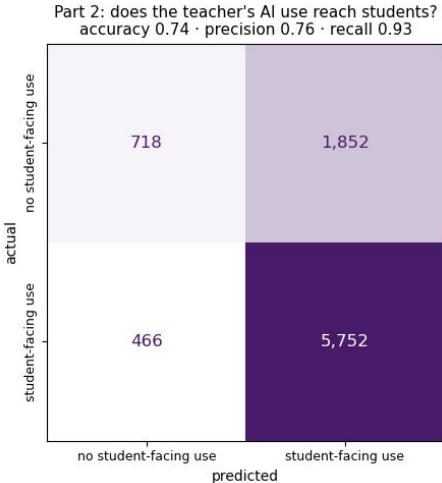


RQ2

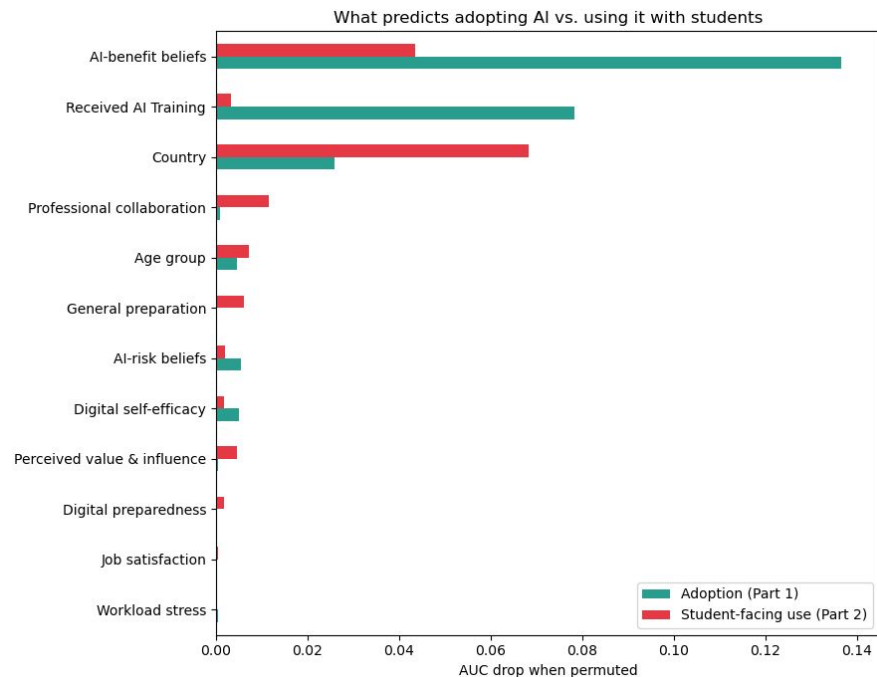
Model comparison — Part 2: Student-facing use grouped 5-fold CV

Model	CV AUC	SD
LogisticRegression	0.753	±0.006
GradientBoosting	0.749	±0.008
RandomForest	0.740	±0.004

Logistic ties gradient boosting (overlapping SDs) — the signal is mostly additive.



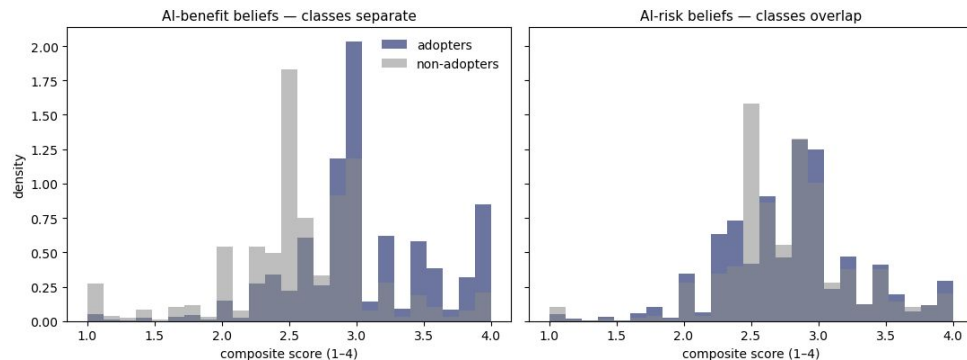
Feature Importance



Perceived usefulness (AI benefit beliefs) beats everything including training and country

Is benefit belief the strongest predictor because of circular two-way effect for AI use?

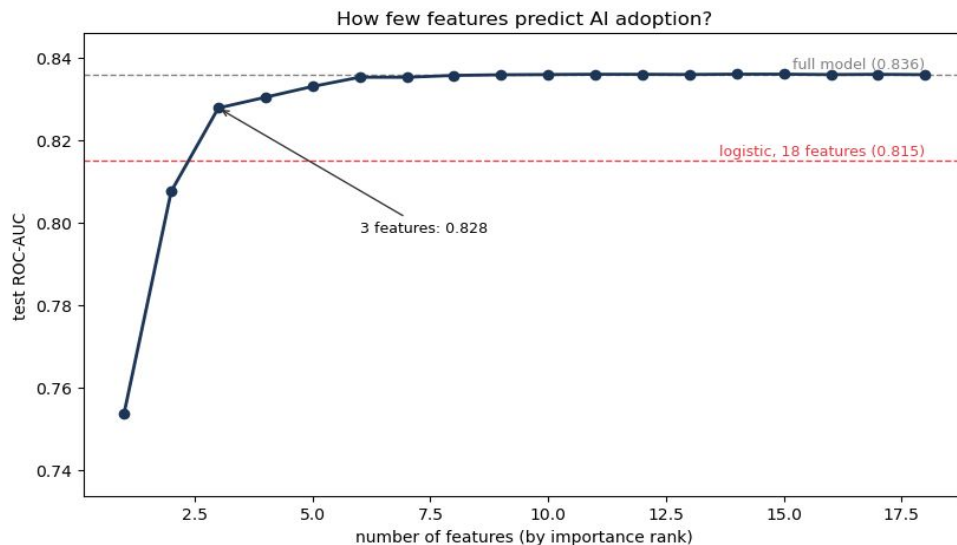
We show that risk beliefs and benefit beliefs come from the same battery, benefit beliefs separate adopters from non-adopters; risk beliefs don't.



Benefit beliefs separate adopters from non-adopters; risk beliefs don't.

Three questions are enough

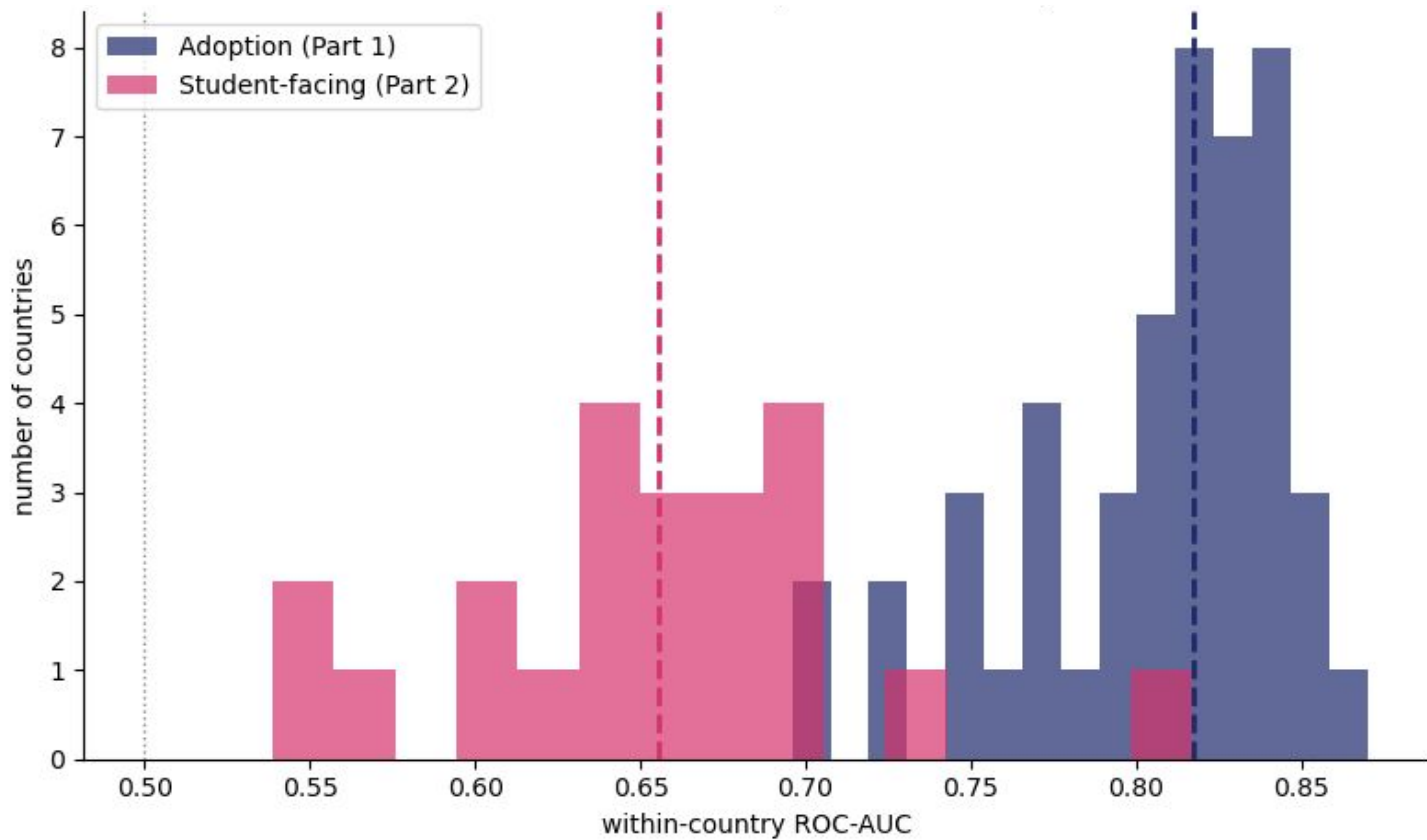
A policymaker doesn't need the full TALIS questionnaire to identify likely adopters. Three questions are enough to capture nearly all the predictable signal. 1) Does the teacher think AI is useful? 2) Have they had AI training? and 3) Which country are they in?



Each feature's marginal contribution

#	Feature added	AUC	Gain
1	AI-benefit beliefs	0.754	+0.254
2	Received AI Training	0.808	+0.054
3	Country	0.828	+0.020
4	AI-risk beliefs	0.830	+0.003
5	Digital self-efficacy	0.833	+0.003
6	Age group	0.835	+0.002

*Full model = 0.836 on this train/test split; the headline 0.833 ± 0.001 averages 10 random splits.



Within countries, adoption is predictable, depth is not.

Results

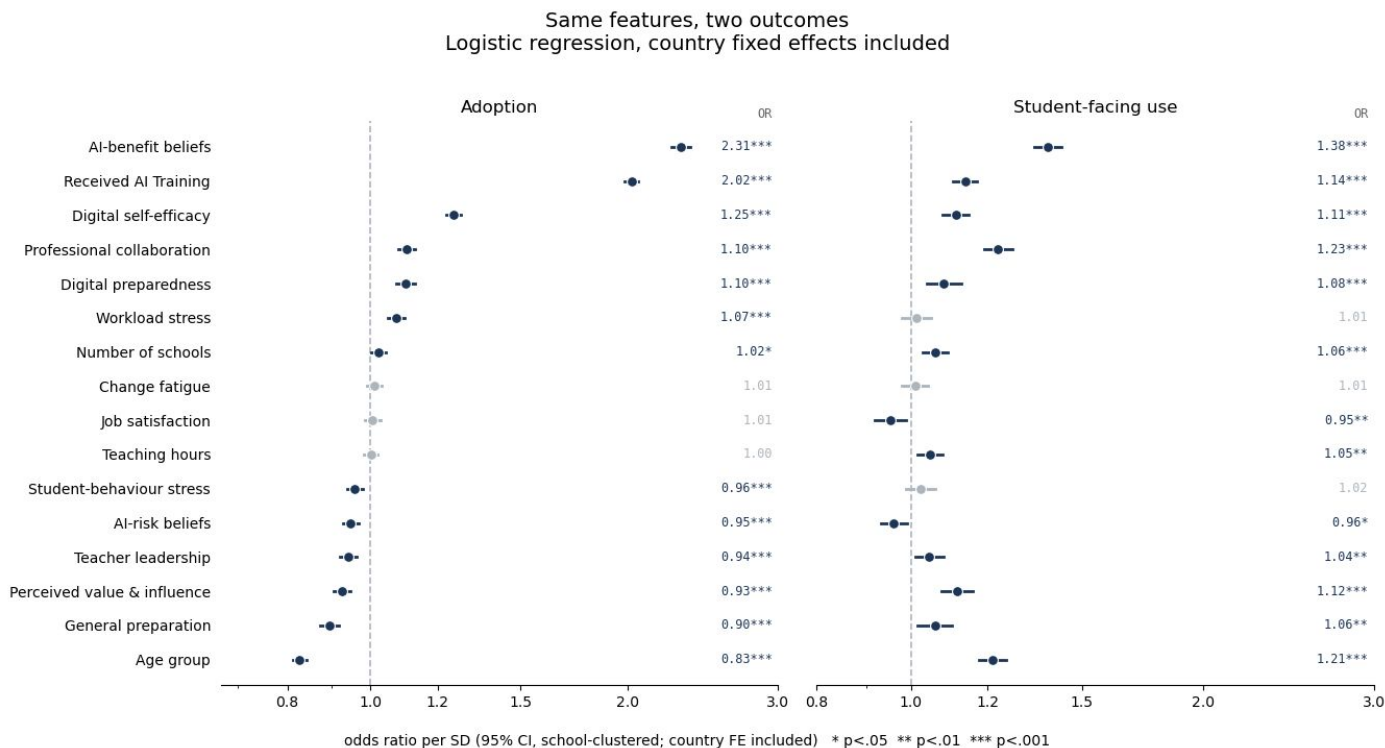
AUC
0.833
Adoption

vs

0.749
Student-facing use

Student-facing use is predictable, but adoption is easier to predict.

Same predictors + country fixed effects; readiness factors explain adoption more strongly than depth of use.



Conclusion

1. Why teachers adopt AI

Adoption is led by perceived usefulness (OR 2.31) and AI training (OR 2.02). Three constructs recover 0.827 of the 0.835 full model

2. Why that doesn't reach students

Under an identical pipeline, discrimination falls to 0.75 and the structure inverts: training's contribution collapses ($+0.068 \rightarrow +0.003$) and country alone (0.698) beats all 17 individual features combined (0.682), a pattern that holds within countries.

3 · Country is a bound, not an explanation

Country absorbs signal from unmeasured context such as countries AI policy, school infrastructure, norms, quantifying what individual data cannot explain. Decomposing it needs system-level measures.

Limitations

Survey Design

AI use was self-reported over the previous 12 months and may be affected by recall or social desirability bias. AI-adoption item appeared only on Form A, administered to one-third of teachers findings so estimates describe the analytic sample rather than national populations.

Survey Weights

Weighted and unweighted results were generally similar, although upper secondary estimates were more weight sensitive.

School block context omitted by necessity

School level variables were unavailable across all participating systems. As a result, country effects may capture differences in school policy, infrastructure, access, and national context that cannot be separated.

Predictive, not causal inference

TALIS 2024 is cross-sectional, and beliefs and AI use were measured at the same time. Therefore, the direction of the relationship cannot be established, and reverse causality is possible.

References

Kaufman, J. H., Woo, A., Eagan, J., Lee, S., & Kassan, E. B. (2025). *Uneven adoption of artificial intelligence tools among US teachers and principals in the 2023-2024 school year*. RAND.

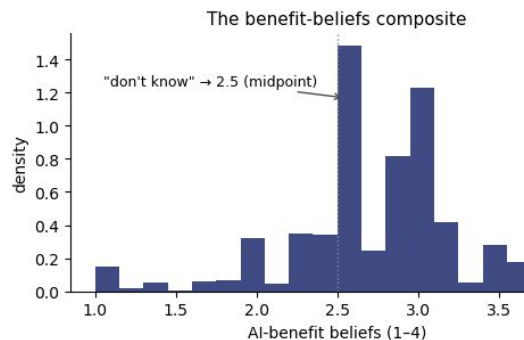
Organisation for Economic Co-Operation and Development (OECD). (2025). Results from TALIS 2024: The state of teaching.

Tripathi, T., Sharma, S. R., Singh, V., Bhargava, P., & Raj, C. (2025, August). Teaching and learning with AI: a qualitative study on K-12 teachers' use and engagement with artificial intelligence. In *Frontiers in Education* (Vol. 10, p. 1651217). Frontiers.

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Appendix

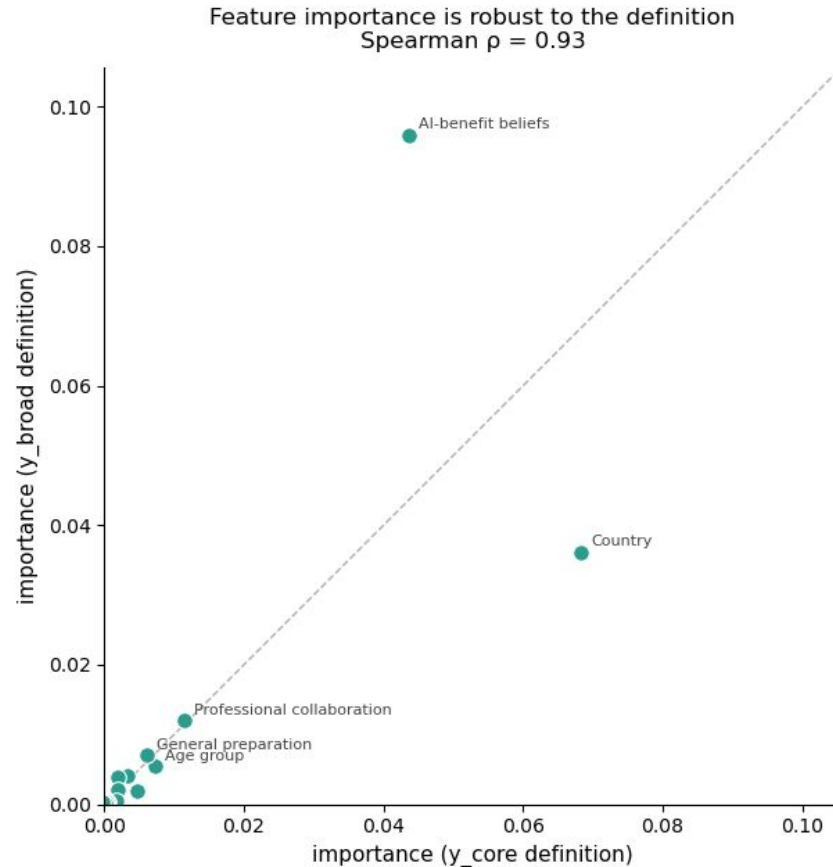
Survey items and belief-composite construction



Item	Item Description	Role
TT4G37A	Assess or mark student work	student-facing (outcome)
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TT4G37C	Generate lesson plans or activities	teacher-facing (descriptive)
TT4G37F	Generate text for feedback or parent communication	teacher-facing (descriptive)

Item	Item Description	Composite
TT4G35A	Helps teachers write/improve lesson plans	ai_benefit_mean
TT4G35B	Enables adaptation of learning material	ai_benefit_mean
TT4G35C	Supports students individually	ai_benefit_mean
TT4G35D	Supports students with special education needs	ai_benefit_mean
TT4G35E	Automates administrative tasks	ai_benefit_mean
TT4G35F	Enables misrepresentation of others' work	ai_risk_mean
TT4G35G	Produces inappropriate/incorrect recommendations	ai_risk_mean
TT4G35H	Amplifies biases	ai_risk_mean
TT4G35I	Jeopardises privacy or security	ai_risk_mean
TT4G35J	Suggests unsuitable pedagogical approaches	ai_risk_mean

RQ2 robustness: Testing whether RQ2 is robust to the outcome definition (adding TT4G37D/E)



Train/test balance: Countries and Features

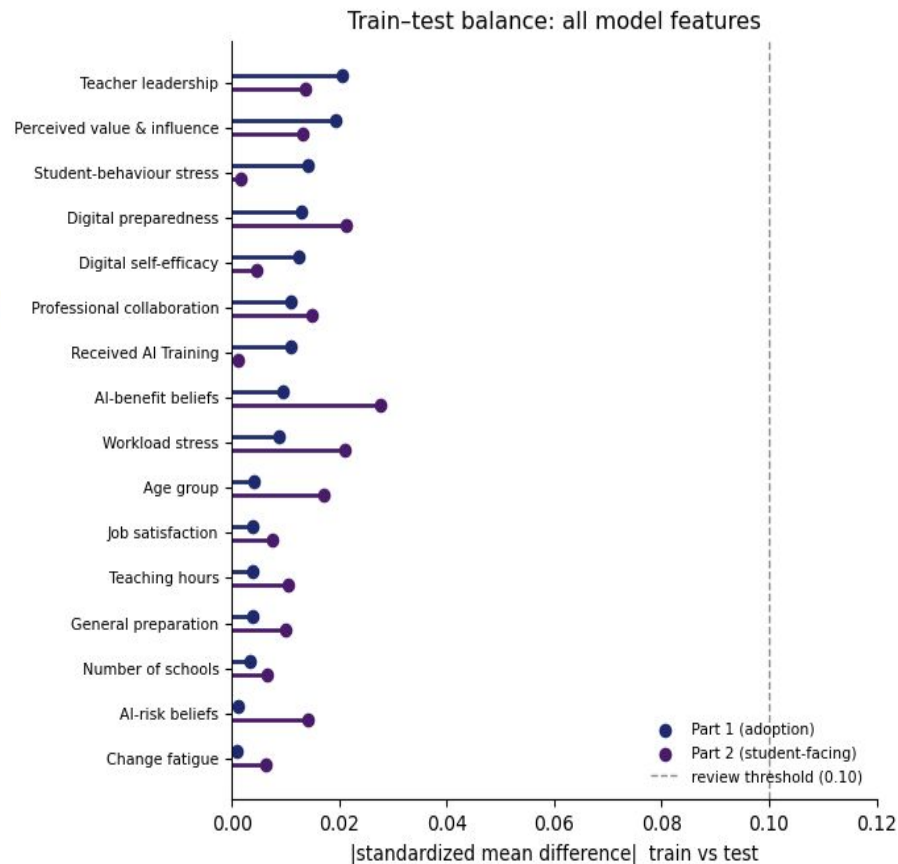
Country representation across the grouped split

Part 1 — adoption

country	train share	test share	diff
TUR	5.4%	4.5%	0.88%
SAU	3.6%	2.9%	0.72%
ESP	6.4%	7.0%	0.59%
ARE	5.1%	4.5%	0.54%
PRT	3.1%	2.5%	0.53%
BHR	1.0%	1.4%	0.40%
max (all)			0.88%

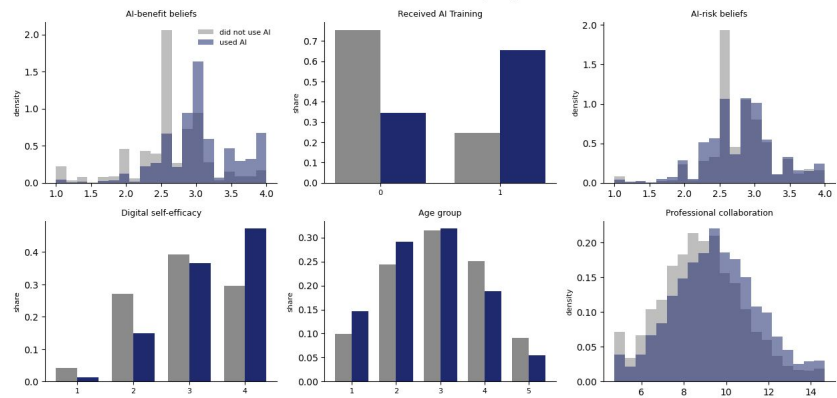
Part 2 — student-facing

country	train share	test share	diff
BRA	2.7%	3.2%	0.49%
VNM	3.0%	2.6%	0.39%
SVN	3.4%	3.0%	0.39%
ESP	5.2%	5.5%	0.34%
SAU	3.9%	3.6%	0.28%
AUS	2.9%	2.6%	0.28%
max (all)			0.49%



Feature distributions by outcome (Part 1: adoption & Part 2: student-facing use)

Part 1 — feature distributions by adoption



Part 2 — feature distributions by student-facing use

